

Quark Mixing from Hyperbolic Geometry: The CKM Matrix from a Compact Arithmetic 3-Manifold

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Abstract

We show that the CKM quark mixing matrix is encoded in the holonomy geometry of the compact arithmetic hyperbolic 3-manifold $M_{\text{CKM}} = m006(-5, 2)$ (`OrientableClosedCensus`[43], volume 2.029, $H_1 = \mathbb{Z}/5$). The symmetric Gaussian overlap construction with word triple $\{\text{aaB}, \text{AbA}, \text{AAb}\}$ and smearing parameter $\sigma = \frac{3}{2} \log \sqrt{13/5}$ achieves Frobenius fitness 0.016482 against PDG 2024, with Cabibbo angle error 0.19% and $p < 0.002$ against 50,000 Haar-random unitary matrices. The Jarlskog invariant vanishes ($J = 0$) at leading order due to a homological class collision: $[\text{AbA}] = [\text{AAb}]$ in $H_1(M_{\text{CKM}}; \mathbb{Z}) = \mathbb{Z}/5$. The invariant trace field of M_{CKM} is a degree-10 number field with defining polynomial $x^{10} - 7x^8 - 4x^7 + 17x^6 + 14x^5 - 18x^4 - 14x^3 + 8x^2 + 3x - 1$, discriminant -271488204251 , and signature $(8, 1)$, confirmed by SnapPy at 500-bit precision. Its predominantly real character (8 real places, 1 complex pair) explains geometrically why the quark sector has suppressed CP violation relative to the lepton sector. All six quark mass ratios m_f/m_e are approximated by norms in $\mathbb{Z}[\sqrt{17}]$ to within 5.7% (permutation null-test $p < 0.002$).

1 Introduction

The CKM quark mixing matrix [1] presents a striking contrast to the PMNS lepton matrix: the CKM angles are small, CP violation is suppressed, and the pattern is close to the identity. In the Hyperbolic Flavor Geometry (HFG) framework [3], these features are encoded in the geometry of a specific compact arithmetic hyperbolic 3-manifold, $M_{\text{CKM}} = m006(-5, 2)$.

The present paper has four main results.

- (i) The symmetric Gaussian overlap construction reproduces the CKM matrix with fitness 0.016482 and Cabibbo angle error 0.19%.
- (ii) The Jarlskog invariant $J = 0$ follows from the word triple geometry, not from any assumption.
- (iii) The invariant trace field of M_{CKM} is a degree-10 number field with signature $(8, 1)$, confirmed computationally.
- (iv) The smearing parameter is $\sigma = \frac{3}{2} \log \sqrt{13/5}$ (exact value derived from the WRT slope norm theorem).

All computations use SnapPy [10] at 150-bit precision.

2 The Manifold

2.1 Basic invariants

Manifold	Census	Slope	Vol	H_1	CS
M_{CKM}	43	$(-5, 2)$	2.029	$\mathbb{Z}/5$	≈ -0.114

$M_{\text{CKM}} = m006(-5, 2)$ is the Dehn filling of $m006$ (the figure-8 knot sister manifold) with slope $(-5, 2)$. The first homology $H_1 = \mathbb{Z}/5$ is inherited from the cusped parent: $m006$ already has $H_1(m006) = \mathbb{Z}/5 \oplus \mathbb{Z}$; the filling kills the $\mathbb{Z}$ factor while preserving the $\mathbb{Z}/5$. This stands in contrast to M_{PMNS} , where the $\mathbb{Z}/5$ is created by surgery on $m003$ (whose cusped $H_1 = \mathbb{Z}$ has no torsion).

2.2 Invariant trace field

Proposition 1 (CKM invariant trace field). *The invariant trace field of M_{CKM} is the number field K with defining polynomial*

$$p(x) = x^{10} - 7x^8 - 4x^7 + 17x^6 + 14x^5 - 18x^4 - 14x^3 + 8x^2 + 3x - 1, \quad (1)$$

discriminant $-271488204251 = -11 \times 239 \times 103266719$, signature $(8, 1)$, and Galois group S_{10} .

Proof. Computed via SnapPy’s `invariant_trace_field_gens()` and Sage’s `find_field(500, 40, optimize=True)`. The result is stable at 200, 300, and 500 bits of precision. Both `trace_field_gens()` and `invariant_trace_field_gens()` return the same polynomial, confirming trace field = invariant trace field. $\square$

Remark 2. Earlier papers in the HFG series claimed the invariant trace field of M_{CKM} was $\mathbb{Q}(\sqrt{17})$. This is incorrect: real quadratic fields have signature $(2, 0)$ (zero complex places) and cannot be the invariant trace field of an arithmetic hyperbolic 3-manifold, which requires exactly one complex place. The correct field is the degree-10 field (1). Note that $\mathbb{Q}(\sqrt{17})$ does not embed in this field.

2.3 Geometric CP suppression

The signature $(8, 1)$ of the ITF means 8 of the 9 Archimedean places are real. By the MacLachlan–Reid theorem [8] (Chapter 3), holonomy elements with traces in the predominantly real ITF are predominantly hyperbolic (real eigenvalues) rather than loxodromic (complex eigenvalues). Purely hyperbolic holonomy gives zero twist angles and hence $J_{\text{CKM}} = 0$ at leading order. This is the geometric explanation of CP suppression in the quark sector: the degree-10 trace field is “more real” (8 real places) than the PMNS trace field (0 real places, since $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\sqrt{-3})$ is imaginary quadratic with signature $(0, 1)$).

2.4 The cover prime and manifold uniqueness

Proposition 3 (Discriminant and cover prime). *The prime $11 = L_5$ (fifth Lucas number) divides $\text{disc}(\text{ITF}(M_{\text{CKM}})) = -11 \times 239 \times 103266719$.*

This connects arithmetically to the topological cover prime: the degree-5 covering space of M_{PMNS} introduces torsion prime 11 [4], and 11 also ramifies in the ITF of M_{CKM} . The prime $11 = \det[(-2, 3), (-5, 2)]$ (the Farey determinant of the two HFG filling slopes) thus appears in both the topology and the arithmetic of both manifolds.

A scan of $m006(p, q)$ fillings confirms that $M_{\text{CKM}} = m006(-5, 2)$ is the unique filling with both $H_1 = \mathbb{Z}/5$ and ITF signature $(8, 1)$: the next filling $m006(-6, 1)$ has signature $(8, 2)$ and is not arithmetic.

3 Mixing Matrix Construction

3.1 Setup

For each geodesic word w , the holonomy $\rho(w) \in \text{PSL}(2, \mathbb{C})$ is computed via SnapPy's `polished_holonomy()` at 150-bit precision. The axis direction $\hat{n}(w) \in S^2$ is extracted from the Pauli decomposition of $\log \rho(w)$. The Gaussian overlap kernel with smearing parameter $\sigma = \frac{3}{2} \log \sqrt{13/5}$ gives

$$K_{ij} = \exp\left(-\frac{\theta_{ij}^2}{2\sigma^2}\right), \quad \theta_{ij} = \arccos(\hat{n}_i \cdot \hat{n}_j).$$

The symmetric matrix K is rendered unitary via QR decomposition.

Remark 4 (Exact smearing parameter). The value $\sigma = \frac{3}{2} \log \sqrt{13/5}$ is derived from the WRT slope norm theorem [6]: $|\text{WRT}_5(M_{\text{PMNS}})|^2 = 13/5 = L_6/|H_1|$, where $L_6 = 13$ is the Gaussian norm of the PMNS filling slope. The golden ratio approximation $\sigma \approx \frac{3}{2} \log \phi$ ($\phi^2 \approx L_6/|H_1|$ to 0.69%) is replaced here by the exact value.

3.2 Word triple and results

Observation 5 (CKM word triple). *The word triple $\{\text{aaB}, \text{AbA}, \text{AAb}\}$ gives geodesic axis angles $\theta_{12} = 48.16^\circ$, $\theta_{13} = 77.48^\circ$, $\theta_{23} = 68.43^\circ$, forming a spherical triangle with excess $13.1^\circ \approx \theta_C$ (Cabibbo angle). With $\sigma = \frac{3}{2} \log \sqrt{13/5}$ and QR orthogonalization:*

Angle	HFG	PDG 2024
θ_{12} (Cabibbo)	13.03°	13.05° (0.19%)
θ_{13}	0.201°	0.201° (< 0.1%)
θ_{23}	2.36°	2.35° (0.4%)

Frobenius fitness $\mathcal{F} = 0.016482$ against PDG 2024. An updated triple $\{\text{abbAbbaB}, \text{Ab}, \text{B}\}$ achieves fitness 0.009078; the canonical triple is retained here for consistency with the companion papers [3].

4 Jarlskog Invariant

Theorem 6 ($J = 0$ from homology). *For the word triple $\{\text{aaB}, \text{AbA}, \text{AAb}\}$, the words AbA and AAb represent the same class $[2] \in H_1(M_{\text{CKM}}; \mathbb{Z}) = \mathbb{Z}/5$. In the QR construction, identical homology classes force the imaginary part of the relevant 3×3 determinant to vanish, giving $J = 0$ exactly at leading order.*

Remark 7. The experimentally measured Jarlskog invariant is $|J| \approx 3 \times 10^{-5}$. The geometric construction yields $J = 0$ at zeroth order; the observed non-zero value must arise from higher-order corrections (e.g., quaternion algebra involution or perturbative flavor mixing).

This provides a geometric explanation of quark CP suppression independent of the trace field argument.

5 Quark Mass Encoding

The six quark mass ratios m_f/m_e are approximated by norms in $\mathbb{Z}[\sqrt{17}]$ (the ring of integers of $\mathbb{Q}(\sqrt{17})$, with norm form $|a^2 - 17b^2|$):

Quark	m_f/m_e (PDG)	Norm witness	Error
u	4.24	$ 2^2 - 17 \cdot 0^2 = 4$	5.7%
d	9.14	$ 3^2 - 17 \cdot 0^2 = 9$	1.5%
s	182.78	$ 14^2 - 17 \cdot 1^2 = 179$	2.1%
c	2495.1	$ 29^2 - 17 \cdot 14^2 = 2491$	0.17%
b	8180.0	$ 117^2 - 17 \cdot 18^2 = 8181$	0.012%
t	338082	$ 139^2 - 17 \cdot 145^2 = 338104$	0.0065%

Five of five non-up quarks have witnesses within 2.1%. Permutation null-test $p < 0.002$ against 50,000 random mass ratios.

Remark 8 (Status of the quark mass ring). The ring $\mathbb{Z}[\sqrt{17}]$ gives empirically excellent witnesses, but $\mathbb{Q}(\sqrt{17})$ does not embed in the confirmed degree-10 ITF of M_{CKM} . The algebraic origin of the $\mathbb{Z}[\sqrt{17}]$ norm structure in terms of the degree-10 field is an open problem.

6 Conclusions

We have shown that the CKM matrix arises from the holonomy geometry of $M_{\text{CKM}} = m006(-5, 2)$ via the symmetric Gaussian overlap construction with the exact smearing parameter $\sigma = \frac{3}{2} \log \sqrt{13/5}$. The Jarlskog invariant $J = 0$ follows from homological class collision. The ITF of M_{CKM} is a degree-10 field with signature $(8, 1)$, whose predominantly real character geometrically suppresses CP violation. The prime $11 = L_5$ appears in both the topology (Farey determinant) and the arithmetic (discriminant of the ITF), connecting both HFG manifolds through the modular curve $X_0(11)$ [5].

Data Availability

Scripts at <https://github.com/drmlgentry/hyperbolic-flavor-scan>.

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